

Unit 4: Magnetism II

- 1) a. Into the page
b. Right (positive x-direction)
c. Out of the page

- 2) a. Clockwise current
- If left closed for a long time = no current
- When opened = counterclockwise current

b. Closed: clockwise
Long time closed: No current
Opened = counterclockwise

c. Closed: clockwise
Long time closed: no current
Opened = counterclockwise

3. a). $\mathcal{E} = \frac{\Delta \Phi_B}{\Delta t}$

$$A = \pi r^2 = \pi (0.0110)^2 = 3.801 \cdot 10^{-4} \text{ m}^2$$

$$\Phi_B = 2 \cdot 3.801 \cdot 10^{-4} = 7.602 \cdot 10^{-4} \text{ Wb}$$

$$\mathcal{E} = \frac{7.602 \cdot 10^{-4}}{.250} = 3.04 \cdot 10^{-3} \text{ V}$$

$$\mathcal{E} = 3.04 \text{ mV}$$

$$b) I = \frac{E}{R} = \frac{3.04 \cdot 10^{-3}}{.0100} = .304 A$$

$$= .304 A$$

$$c) P = I^2 R = (.304)^2 \cdot .0100 = .000924 W$$

$$P = .924 mW$$

$$4. a) N = \frac{E_{peak}}{BAW}$$

$$A = .0160 \cdot .0300$$

$$= 3 \times 10^{-4} m^2$$

$$N = \frac{18.0}{3 \cdot 360 \cdot 1875 \cdot 10^{-4}} = \frac{18}{.675} = 26.67$$

$$N = 27 \text{ turns}$$

$$b) T = \frac{2\pi}{1875} = \frac{6.2832}{1875} = 3.35 \cdot 10^{-3} s$$

$$T = 3.35 ms$$

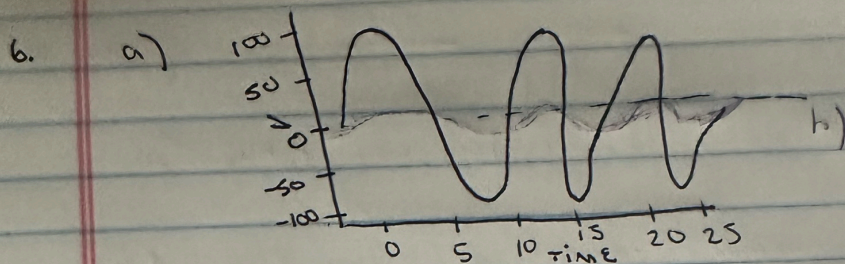
$$5. a) \frac{V_p}{V_s} = \frac{N_p}{N_s}$$

$$\frac{N_p}{N_s} = \frac{240}{120} = 2$$

$$= 2$$

$$b) \frac{I_p}{I_s} = \frac{120}{240} = 1/2$$

$$\frac{I_{in}}{I_{out}} = 1/2 \text{ or } \frac{I_{out}}{I_{in}} = 2$$



$y = V$ ranging from $-120V$ - $+120V$

$x = ms(time)$, 3 full cycles

• Sine wave: $V_0 = 120V$

$$\omega = 240\pi \text{ rad/s} \rightarrow f = 120\text{Hz}, T = \frac{1}{f} = 8.33\text{ms}$$

b) c) $v(t) = 120 \sin(240\pi t) = 0 \rightarrow t = n/240, n=0,1,2,\dots$
 $t = 0, 4.17\text{ms}, 8.33\text{ms}, 12.5\text{ms}, \dots$

7) $\mathcal{E} = L \frac{\Delta I}{\Delta t}$

$$\Delta t = \frac{2 \cdot 10^{-3} \cdot 100}{500} = \frac{200 \cdot 10^{-4}}{500} = 4 \times 10^{-7} \text{ s}$$

$$\Delta t = .400\mu\text{s}$$

$$8. \quad a) \quad \mathcal{E} = 25 \cdot \frac{100}{0.0800} = 25 \cdot 1250 = \underline{31,250 \text{ V}}$$

$$b) \quad E = \frac{1}{2} \cdot 23 (100)^2 = 12.5 \cdot 10,000 = \underline{125,000 \text{ J}}$$

$$c) \quad \frac{125,000}{0.0800} = 1,562,500 \text{ W} \quad \text{or} \quad \underline{1.56 \cdot 10^6 \text{ W}}$$

$$? \quad d) \quad L = \mu_0 \frac{N^2 A}{l} \rightarrow N = \sqrt{\frac{L l}{\mu_0 A}}$$

$$L = 25 \text{ H}$$

$$\text{Cross-sec: } A = .01 \text{ m}^2$$

$$l = .5 \text{ m}$$

$$N = 31,539 \text{ turns}$$

$$e) \quad X_L = 2\pi f L = 2\pi (10,000)(25) = 500,000 \pi = 1.57 \cdot 10^6 \Omega \\ = 1.57 \text{ M}\Omega$$

$$9. \quad a) \quad L = (20 \cdot 10^{-9}) (5 \cdot 10^6) = \underline{1.100 \text{ H}}$$

$$b) \quad T = 1 \text{ ns} \quad R = \frac{.100}{1.00 \cdot 10^{-9}} = 1.00 \cdot 10^8 \Omega = \underline{100 \text{ M}\Omega}$$

$$c) \quad \frac{I(t)}{I_0} = e^{-3/20} = e^{-.15} = .8607$$

$$= 86.1\%$$

10. a) $V_{in} - V_R - V_C = 0 \rightarrow V_{in} = IR + V_C$

b) across: $V_{out} \rightarrow V_C \rightarrow V_{out} = Q/C$

$V_{out} = I \cdot Z_C = I \cdot \frac{1}{j\omega C}$

c) $\frac{V_{out}}{V_{in}} = \frac{Z_C}{R + Z_C} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$

d) $\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (\omega RC)^2}} = \frac{1}{\sqrt{1 + 100^2}} = \frac{1}{\sqrt{10001}} = .009999$
 $= -10\%$

$\left| \frac{V_{out}}{V_{in}} \right| = \frac{1}{\sqrt{1 + (\omega)^2}} = \frac{1}{\sqrt{1.01}} = .995$
 $= -99.5\%$

11.

a) $f_0 = \frac{1}{2\pi\sqrt{10^{-2} \times 10^{-6}}} = \frac{1}{2\pi \times 10^{-4}} = \frac{1}{6.2832 \times 10^{-4}} = 1592 \text{ Hz}$

b) $f = .1 f_0 = 159.2 \text{ Hz}$

$X_L = 2\pi \cdot 159.2 \cdot .01 = 10.0 \Omega$

$Z = \sqrt{100^2 + (10 - 100)^2} = \sqrt{100^2 + 90^2} = 99.5 \Omega$

$V_C = \frac{1}{2\pi \cdot 159.2 \cdot 10^{-6}} = 10.0 \Omega$

$Z = \sqrt{100^2 + (100 - 10)^2} = \sqrt{100^2 + 90^2} = 99.5 \Omega$

12. a) $I_{rms} = \frac{V_{rms}}{Z}$

$$I_{rms} = \frac{120}{995} = .121A$$

b) $f_0 = 1592Hz$

$$Z = 995\Omega @ .1f_0 \text{ \& } 10f_0$$

$$I_{rms} = .121A, P_{rms} = 1.47W$$

$$P_{rms} = (.121)^2 \cdot 100 = 1.47W$$

13. carrier (v) $f = 1.4MHz$

$$A_{v,0} = f_m = 10kHz$$

1. Carrier $f = f_c = 1.4MHz$

2. Upper sideband: $f_c + f_m = 1.4MHz + 10kHz = 1.410MHz$

3. Lower = $1.4MHz - 10kHz = 1.390MHz$

$$1.390, 1.400, 1.410$$

Units.

1) a) $\frac{\Delta B}{\Delta t} = \frac{2 \cdot 10^{-7}}{2 \cdot 10^{-6}} = .100A$

$$B = \frac{(4\pi \cdot 10^{-7})(.100)}{2\pi \cdot .01} = \frac{4 \cdot 10^{-8}}{.01} = 4 \cdot 10^{-6}T$$

$$= 4.00\mu T$$

b) Displacement current (changing electric field between plates)

$$2 \quad a) \quad \frac{3 \times 10^8 \cdot 10 \cdot 10^{-6}}{2} \cdot \frac{1}{1500 \text{ m}}$$

$$b) \quad R = \left(\frac{1 - 1.78}{1 + 1.78} \right)^2 = \left(\frac{-0.78}{2.78} \right)^2 = (.280)^2 = .0784$$

$$\text{or } 7.84\%$$

3.

$$a) \quad \frac{1000}{.0010} = 1.0 \cdot 10^6 \text{ W/m}^2$$

$$b) \quad I_1 \left(\frac{r_1}{r_2} \right)^2 = (1.0 \cdot 10^6) \left(\frac{1}{2} \right)^2 = 1.0 \cdot 10^6 / 4 = \underline{2.5 \cdot 10^5 \text{ W/m}^2}$$

$$4. ? a) \quad \theta_3 = \theta_1 \quad \text{if } n_1 = n_3 ?$$

$$b) \quad \frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{12} - \frac{1}{8} = \frac{2-3}{24} = \frac{1}{24} \rightarrow d_i = -24 \text{ cm}$$

$$M = \frac{d_i}{d_o} = \frac{-24}{8} = \underline{-3.0}$$

$$5. \quad n \mu x = (3.3 \cdot 10^{28}) (200 \cdot 10^{-28}) (.01) = 6.6$$

$$\frac{1}{I_0} = e^{-6.6} = .00136 \rightarrow .136\% \text{ , blocks } 99.86\%$$

$$b. a) \text{ dose} = \frac{250 \cdot 10^3}{60} \cdot 100 = \underline{417 \text{ rad}}$$

$$b) \frac{250 \cdot 10^3}{2} \cdot 100 = \underline{12.5 \text{ rad}}$$

$$c. \text{ dose} \cdot RBE = \left(\frac{12.5}{100} \right) \cdot 1 = \underline{0.125 \text{ Sv}}$$

d. Yes = moderate radiation dose which can increase cancer risk